Q51

```c
#include<stdio.h>
int main()
{
    int size, i,j;
    printf("\n enter amount of numbers:");
    scanf("%d", &size);
    int num[size];
    for (i=0; i<size; i++)
    {
        printf("\n enter a number: ");
        scanf("%d", &num[i]);
    }
    printf("numbers:");
    for (i=0; i<size; i++)
    {
        printf("\t %d", num[i]);
    }
    int rev[size];
    for (i=0; i<size; i++)
    {
        rev[size-1-i] = num[i];
    }
    for (i=0; i<size; i++)
    {
        num[i] = rev [i];
    }
    printf("\n reversed: ");
    for (i=0; i<size; i++)
    {
        printf("\t%d", num[i]);
    }
}
```

```
enter a number: 4

enter a number: 5

enter a number: 6
umbers:         1       2       3       4       5       6
reversed:       6       5       4       3       2       1
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```

Q52

```c
#include<stdio.h>
int main()
{
    int matrix[3][3],max,min,i,j;
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            printf("enter an element:");
            scanf("%d", &matrix[i][j]);
        }
    }
    printf("\nmatrix\n");
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            printf("%d\t", matrix[i][j]);
        }
        printf("\n");
    }
    max=matrix[0][0];
    min=matrix[0][0];
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            if(matrix[i][j]>max)
            {
                max = matrix[i][j];
            }
            else if(matrix[i][j]<min)
```

```
            {
                min = matrix[i][j];
            }
        }
    }
    printf("minimum:%d; maximum:%d", min,max);
}
```

```
enter an element:1
enter an element:2

matrix
12      13      14
15      19      20
21      1       2
minimum:1; maximum:21
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```

Q53

```
#include<stdio.h>
int main()
{
    int matrix[3][3],sum,i,j;
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            printf("enter an element:");
            scanf("%d", &matrix[i][j]);
        }
    }
    printf("\nmatrix\n");
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            printf("%d\t", matrix[i][j]);
        }
        printf("\n");
    }
```

```
    for(i=0;i<3;i++)
    {
        sum = 0;
        for(j=0;j<3;j++)
        {
            sum = sum + matrix[i][j];
        }
        printf("\naddition of now no. %d elements: %d", i+1, sum);
    }
}
```

```
matrix
12      13      14
1       2       3
50      60      70

addition of now no. 1 elements: 39
addition of now no. 2 elements: 6
addition of now no. 3 elements: 180
PS E:\intro to prog assignments sem1>
```

Q54

```
#include<stdio.h>
int main()
{
    int matrix[3][3],temp,i,j;
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            printf("enter an element:");
            scanf("%d", &matrix[i][j]);
        }
    }
    printf("\nmatrix\n");
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            printf("%d\t", matrix[i][j]);
```

```
        }
        printf("\n");
    }
    for(i=0;i<3;i++)
    {
        for(j=i;j<3;j++)
        {
             temp = matrix[i][j];
             matrix[i][j] = matrix[j][i];
             matrix[j][i] = temp;
        }
    }
    printf("transpose:\n");
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            printf("%d\t", matrix[i][j]);
        }
        printf("\n");
}
}
```

```
matrix
12      13      14
1       2       3
6       7       9
transpose:
12      1       6
13      2       7
14      3       9
PS E:\intro to prog assignments sem1>
```

Q55

```
#include<stdio.h>
int main()
{       int ma1[3][3],ma2[3][3],ma3[3][3],i,j;
        printf("\nENTER MATRIX 1\n");
        for(i=0;i<3;i++)
        {
            for(j=0;j<3;j++)
```

```
            {
                printf("enter an element:");
                scanf("%d", &ma1[i][j]);
            }
        }
        printf("\nENTER MATRIX 2\n");
        for(i=0;i<3;i++)
        {
            for(j=0;j<3;j++)
            {
                printf("enter an element:");
                scanf("%d", &ma2[i][j]);
            }
        }

        for(i=0; i<3;i++)
        {
            for (j=0;j<3;j++)
            {
                ma3[i][j] = ma1[i][j] + ma2[i][j];
            }
        }

        printf("\nAddition:\n");

        for(i=0; i<3;i++)
        {
            for (j=0;j<3;j++)
            {
                printf("%d\t", ma3[i][j]);
            }
            printf("\n");
        }
}
```

```
ENTER MATRIX 1
enter an element:1
enter an element:2
enter an element:3
enter an element:-1
enter an element:-2
enter an element:-3
enter an element:9
enter an element:8
enter an element:7

ENTER MATRIX 2
enter an element:55
enter an element:33
enter an element:22
enter an element:5
enter an element:6
enter an element:7
enter an element:10
enter an element:20
enter an element:30

Addition:
56      35      25
4       4       4
19      28      37
PS E:\intro to prog assignments sem1>
```

Q56

```c
#include<stdio.h>
int main()
{       int ma1[3][3],ma2[3][3],ma3[3][3],i,j;
        printf("\nENTER MATRIX 1\n");
        for(i=0;i<3;i++)
        {
            for(j=0;j<3;j++)
            {
                printf("enter an element:");
                scanf("%d", &ma1[i][j]);
            }
        }
        printf("\nENTER MATRIX 2\n");
        for(i=0;i<3;i++)
        {
            for(j=0;j<3;j++)
            {
                printf("enter an element:");
                scanf("%d", &ma2[i][j]);
            }
        }
```

```
        for(i=0; i<3;i++)
        {
            for (j=0;j<3;j++)
            {
                ma3[i][j] = ma1[i][j] - ma2[i][j];
            }
        }

        printf("\nSubtraction:\n");

        for(i=0; i<3;i++)
        {
            for (j=0;j<3;j++)
            {
                printf("%d\t", ma3[i][j]);
            }
            printf("\n");
        }
}
```

```
ENTER MATRIX 1
enter an element:45
enter an element:67
enter an element:98
enter an element:21
enter an element:12
enter an element:11
enter an element:1
enter an element:2
enter an element:3

ENTER MATRIX 2
enter an element:21
enter an element:8
enter an element:3
enter an element:5
enter an element:121
enter an element:200
enter an element:-10
enter an element:2
enter an element:46

Subtraction:
24      59      95
16      -109    -189
11      0       -43
PS E:\intro to prog assignments sem1>
```

Q57

```c
#include<stdio.h>
int main()
{       int ma1[3][3],ma2[3][3],ma3[3][3],i,j,temp,k;
        printf("\nENTER MATRIX 1\n");
        for(i=0;i<3;i++)
        {
            for(j=0;j<3;j++)
            {
                printf("enter an element:");
                scanf("%d", &ma1[i][j]);
            }
        }
        printf("\nENTER MATRIX 2\n");
        for(i=0;i<3;i++)
        {
            for(j=0;j<3;j++)
            {
                printf("enter an element:");
                scanf("%d", &ma2[i][j]);
            }
        }

        for(i=0; i<3;i++)
        {

            for (j=0;j<3;j++)
            {
                ma3[i][j] = 0;
                for(k=0;k<3;k++)
                {
                ma3[i][j] = ma3[i][j] + ma1[i][k]*ma2[k][j];
                }
            }
        }

        printf("\nMatrix multiplication [Matrix 1].[Matrix 2] :\n");

        for(i=0; i<3;i++)
        {
            for (j=0;j<3;j++)
            {
                printf("%d\t", ma3[i][j]);
```

```
            }
            printf("\n");
        }
}
```

```
ENTER MATRIX 1
enter an element:12
enter an element:13
enter an element:14
enter an element:15
enter an element:16
enter an element:17
enter an element:18
enter an element:19
enter an element:20

ENTER MATRIX 2
enter an element:1
enter an element:2
enter an element:3
enter an element:4
enter an element:5
enter an element:6
enter an element:7
enter an element:8
enter an element:9

Matrix multiplication [Matrix 1].[Matrix 2] :
162     201     240
198     246     294
234     291     348
PS E:\intro to prog assignments sem1>
```

Q58

```
#include<stdio.h>
int main()
{
    int ma1[3][3]= {{1,3,5},{2,4,6},{10,11,12}};
    int key;
```

```
    int i,j,flag=0;
    printf("enter the number to be searched:");
    scanf("%d",&key);
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            if(ma1[i][j] == key)
            {
                flag = 1;
            }
        }
    }
    if (flag==1)
    {
        printf("found");
    }
    else
    {
        printf("not found");
    }
}
```

```
PS E:\intro to prog assignments sem1\lab6> ./a.exe
enter the number to be searched:6
found
PS E:\intro to prog assignments sem1\lab6>
```

Q59

```
#include<stdio.h>
int main()
{
    int array[3][3],i,j,key,occ=0;
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            printf("\nenter an element:");
            scanf("%d",&array[i][j]);
        }
```

```
    }
    printf("\nenter the number to be searched:");
    scanf("%d", &key);
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
           if(array[i][j]==key)
           {
            occ++;
           }
        }
    }

if(occ>0)
{
    printf("\nelement %d occurs %d times", key,occ);
}
else
{
    printf("\nelement does not exist");
}
}
```

```
enter an element:1

enter an element:1

enter an element:1

enter an element:23

enter an element:22

enter an element:12

enter an element:1

enter an element:4

enter an element:5

enter the number to be searched:1

element 1 occurs 4 times
PS E:\intro to prog assignments sem1\lab6>
```